Arabic GPT from Scratch

Decoder-Only Transformer for Arabic Poetry Generation | Built with PyTorch

1. Project Overview

A GPT language model built entirely from scratch using PyTorch — no pretrained weights. Trained on Arabic Wikipedia then fine-tuned on Arabic poetry. The model has 23 million parameters and uses a BPE tokenizer trained on Arabic data.

2. Data

Pretraining

- Source: Arabic Wikipedia XML dump (12.23 GB)
- After cleaning: 50,000 lines, 6.7 MB
- Cleaning: removed URLs, normalized Alef, removed Tatweel & diacritics

Fine-Tuning (SFT)

- Source: arbml/ashaar (HuggingFace) — 254,630 poems
- Used: 500 instruction-response pairs
- Format: { "instruction": "اكتب قصيدة عن ...", "output": "أبيات القصيدة" }

3. Tokenizer

- Algorithm: ByteLevel BPE trained from scratch on Arabic Wikipedia
- Vocabulary size: 16,000 (suited for Arabic morphology)
- Special tokens: <pad> <sos> <eos> <unk>
- Example: "كيميائية مادة الماء" → 11 subword tokens

4. Model Architecture

Component	Value
Parameters	22,960,128 (~23M)
Embed Size	384
Layers	6 Transformer blocks
Attention Heads	6
Context Length	64 tokens

Activation	GELU
Dropout	0.1

Built from scratch: Self-Attention (Q/K/V matrices), Causal Masking, Residual Connections, LayerNorm, Positional Embeddings, Weight Initialization (std=0.02).

5. Training & Optimization

Setting	Pretraining	Fine-Tuning
Optimizer	AdamW	AdamW
Learning Rate	3e-4	1e-4
Epochs	10	20
Batch Size	32	16
Scheduler	CosineAnnealingLR	—
Gradient Clipping	1.0	1.0

Pretraining loss: 2.28 → 0.59 (73.8% improvement). Smooth curve — no overfitting detected.

6. Evaluation

Perplexity

Model	Perplexity
Pretrained (before fine-tuning)	65,697.52
Fine-tuned (after poetry training)	1.45

LLM Judge (Ollama gemma3:4b) — Scale 1-5

Sample	Score	Feedback
الوطن عن قصيدة	3/5	Some recognizable structure but grammatically inconsistent
الأم عن قصيدة	1/5	Very poor linguistic quality with significant grammatical errors
البحر عن قصيدة	1/5	Unclear wording and multiple spelling errors
Average	1.7/5	Model shows early signs of poetry structure but needs improvement

Failure Modes

- Incoherent grammar — model too small (23M vs 117M for GPT-2)
- Context loss after 3-4 verses — context window only 64 tokens
- No consistent rhyme scheme — requires larger model
- Topic drift in longer generations

7. Key Learnings

- Built a complete GPT architecture from scratch — Attention, Residuals, LayerNorm, Positional Embeddings
- Arabic NLP requires special handling: normalization, diacritics removal, right-to-left text
- BPE vocabulary size of 16,000 is appropriate for Arabic morphological richness
- Two-phase training (pretrain then fine-tune) is essential for task-specific language models
- Perplexity is a reliable metric for comparing model performance across training phases
- Small models (23M) can learn language patterns but struggle with complex creative tasks like poetry
- Gradient clipping and cosine LR scheduling are critical for stable transformer training

8. Limitations & Future Work

- Scale to 117M+ parameters for fluent Arabic generation
- Increase context window to 512+ tokens
- Train on 100,000+ poems instead of 500
- Implement beam search for better output quality
- Add rhyme-aware loss function for consistent meter